

Literature Review and Research Proposal Outlines (Research Methods and Professional Practice)

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Topic: Cybersecurity threats in IoT for the Power and Utilities sector (smart grids and SCADA)

1. Introduction

- 1.1 Main goal: Check how using IoT in smart-grid and SCADA setups shifts security risks, and spot controls that cut down real-world dangers.
- 1.2 Readers: Experts in research, power company security groups, rule makers for standards, and government watchers.
- 1.3 Goal and value: Help safe tech upgrades by linking proof of threats to practical fixes and guidelines.
- 1.4 Methodology for locating and selecting literature: Databases—IEEE Xplore, ACM DL, ScienceDirect, IET, NIST, ENISA. Keywords— ‘smart grid security’, ‘SCADA/ICS security’, ‘IEC 62351’, ‘NIST SP 800-82’, ‘AMI cyber security’. Inclusion—2010–2025; peer-reviewed or standards; utility focus.

2. Conceptual foundations

- Layers and areas in smart-grids and SCADA; gaps between IT goals and OT needs.
- Key links and tools: IEC 60870-5-104, DNP3, Modbus/TCP; entry points; back systems for AMI.
- Guide framework: NIST SP 800-82; NISTIR 7628; IEC 62351.

3. Sector technological demands

- Limits in daily work: Keeping safe, steady timing, old gear life, mixed suppliers.
- Forces for change: Adding in solar/wind and electric cars, big growth in AMI, cloud data checks.
- Common wants: Safe data sends, far-off fixes, updates you can check.

4. Applications and threat scenarios

- Hacks on field tools, bad software loads, stolen codes.
- Jumps from substations through wrong remote links.
- AMI data leaks and tricks on power use patterns.
- Messing with time links and power state checks.

5. Analytical framework

- Link risks to key parts (tools, nets, apps, oversight).
- Score fixes by ease in active OT (split areas, safe links, watch tools, rules).

6. Findings

- Good effects from top fixes: Sharp borders, less side jumps, checks on trust points.
- Ongoing issues: Old machines, update gaps, tied to suppliers, staff shortages.

7. Strengths and limitations of existing work

- Pros: Firm rules and growing advice.
- Cons: Rare direct comparisons; thin info on costs and people factors.

8. Discrepancies

- Real use of strong codes at far points.
- How well watch systems work with low band and strict timing.
- Cloud use for quick tasks.

9. Conclusions and recommendations

- Use protocol setups that match IEC 62351.
- Set strict areas and approved talks. • Roll out monitoring fit for OT.
- Map out shifts from old gear.
- Run regular test attacks with safety stops.

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